

Smart Grievance Redressal System

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Abstract—Addressing the grievances of citizens is a tough task for an administration. Classifying and prioritizing the grievances will make the grievance redressal system more efficient. This paper proposes a software application to identify common issues by analysing multiple grievances, so as to provide the administration, with a view of the important issues that require immediate attention. The software application is developed using state of the art NLP technologies to implement text classification, keyword extraction and keyword clustering on the set of grievances.

Index Terms—Text Classification, Keyword Extraction, Keyword Clustering, fastText, Multipartite Rank Algorithm, BERT, ANNOY

I. INTRODUCTION

The administration gets thousands of grievances in a day and it becomes difficult to respond to each one of them. The rate of incoming grievances tend to be high. So attending to individual issues is not practical, especially when the population is huge. Sometimes there are similar complaints or grievance being registered multiple times, which means that the issue needs urgent attention. To the best of our understanding, there is no existing software application to analyze the grievances, in terms of the similarity of grievances and generate a summary.

The proposed software application would determine the authority to which a particular grievance must be directed for redressal using a supervised machine learning model. The application would provide a consolidated view of the gravity of the grievances by the periodic application of a combination of keyword extraction and keyword clustering. This helps each department within the administration to prioritize the issues before solving.

Grievances are fed into the system, which are then directed to a text classifier resulting in being tagged with the department, which must redress the grievance. After the classification, the grievances are fed into a keyword extraction algorithm that tags the classified complaints with their keywords. The obtained result, which contains the grievance, the department concerned with redressing the grievance and keywords will be stored into the database.

It is possible that the same grievance is specified differently. To identify semantically similar text, a routine job, is executed on an hourly basis, which fetches keywords grouped by departments. These keywords are then fed into their respective department-based clustering models. These department specific clustering models are then retrained with the newly obtained keywords. Then these clustering models are used to associate the original keyword with the keywords from new

grievances, within the same department, that are closest in meaning to the former. The results are stored in the database.

With the newly available data of each keyword tagged with their related keywords, we can calculate the total count of grievances by summing frequency of occurrences of a keyword and its related keywords

The paper is organised as follows: A brief review of the relevant literature is presented in the next section. Section III presents the details of the data set. The architecture and implementation of the proposed software application is presented in Section IV. The results of testing the software are presented Section V.

II. LITERATURE SURVEY

The proposed software application involves three modules namely text classification, identifying semantically similar text and aggregating semantically similar text. In this section, a survey of related literature is presented.

Most of the research literature can be broadly segregated into two : pre processing of text before classification and machine learning techniques for text classification. Supervised and unsupervised machine learning techniques have been used for text classification. A review of the machine learning techniques used for classification of text can be found in the paper by Aurangzeb Khan et.al.[1].

One of the important steps in text classification is keyword extraction. Eibe Frank et. al. [2] proposed a Naive Bayes approach for classifying words as keywords.

Rough sets based text classification was proposed by Bao Y. et. al. [3]. The paper proposed applying multiple reducts to improve the performance of kNN.

Bao Y et. al. [4] proposed an approach towards combining multiple kNN classifiers based on different distance functions. The proposed form of kNN algorithm seeks to increase generalization accuracy when compared to the basic kNN algorithm.

R. Mihalcea et. al. [5], [6] applied graph based ranking algorithms for extracting sentences from text. The algorithms discussed were Hyperlinked induced topic search, The positional power function and PageRank algorithm. The algorithms considered the number of hyperlinks to a document. Graph based algorithms represents the document as a semantic graph and determines the importance of a vertex from the information in the entire graph.

Rakesh Gupta et. al. [7] showed that the bag of words classifier performs poorly on noisy short conversations. The

paper discusses extraction of features from different knowledge bases. The features are then combined and pruned based on mutual information.

Davide Magatti et. al. [8] presented a software system for classification of documents based on topic extraction. LDA(Latent Dirichlet Allocation) model was used for Topic Extraction and a specialized version of the Multi-Net Naive Bayes model was used for performing Multi-label document classification.

Takeru Yokoi et. al. [9] proposed a method for extracting topics from large set of documents. They applied Non-Negative Matrix Factorization techniques on small document set and integrated them to extract topics from larger document set.

Aman Kumar Sharma et. al. [10] compared four algorithms namely: ID3, J48, Simple CART and Alternating Decision Tree, for classification of legitimate email messages from spam email message. The results showed clearly that the J48 classifier performed with highest classification accuracy.

Leonidas Akritidis et. al. [11] proposed automatic classification of research articles. They used kNN using a relevance description vector which defines the frequency with which each keyword is related to each research field.

Sonu Yadav et. al. [12] proposed a support vector machine based approach to identify semantic similarity between words and text documents to improve efficiency of search.

Muhammad Shoaib et. al. [13] proposed a framework for clustering news articles using fuzzy c - means. They rank the tweets on the basis of the retweets of a tweet. The approach is based on constructing a graph connecting users, tweets and their genre.

Tomas Mikolov et. al. [14], [15] introduced techniques that can be used for learning word vectors from large data sets of words. Word2Vec representation is created using algorithms namely Continuous Bag-of-Words Model or Continuous Skip-gram Model. Continuous bag of words (CBOW) predicts one word from the words surrounding it. The Skip Gram model uses one word to predict all surrounding words. Skip Gram is much slower than CBOW, but considered more accurate with infrequent words.

Thien Hai Nguyen et. al. [16] proposed a multi-label based topic extraction model for research publications. They used a back-off and claimed improved accuracy over models using only kNN.

Quoc Le and Tomas Mikolov [17] improved on their previous work and developed a numeric representation of documents called doc2vec. An unsupervised algorithm namely paragraph vector was proposed. The algorithm learns features from sentences, paragraphs and documents.

Beakcheol Jang et. al. [18] compared word2vec Convolutional Neural Networks (CNNs) to classify news articles and tweets as related and unrelated ones. The experiment was carried out with CNNs using Continuous Bag of Words(CBOW) and Skip Gram models. The results indicated that CBOW model performed better on news articles while Skip Gram model performed better with respect to tweets.

Sunilkumar P et. al. [19] classified semantic similarity approaches as corpus-based, knowledge based and string-based. The methods included in the survey are Latent Semantic Analysis (LSA), Word Embedding, Word2Vec, GloVe, fastText, Cosine Similarity, Jaccard Similarity, Levenshtein distance, n-gram. They concluded that Knowledge-based and Corpus-based methods are widely used for measuring semantic similarity.

M G Thushara et. al. [20] compared the unsupervised keyword extraction methods namely Brute Force, RAKE (Rapid Automatic Keyword Extraction), TextRank and PositionRank algorithms. While Brute Force Approach considers every word as a possible candidate for keyword, RAKE constructs a keyword score matrix using Word Frequency, Word degree and ratio of word frequency to word degree to determine keyphrases. TextRank uses a graph ranking algorithm like PageRank. PositionRank, which is similar to TextRank, assumes that the phrases which occur in the beginning of the document are more important than the phrases that appear later in the document. PositionRank constructs a graph by connecting two words by an edge, if these words co-occur within a window of w contiguous tokens.

Ikuya Yamada et. al. [21] proposes a text classification model by using entities in a knowledge base. It uses neural networks to perform text classification. This model reduces the ambiguity as each entity is linked to the knowledge base and the context in which the word is used is checked.

Dimo Angelov [22] proposed Top2vec, a method to find topic vectors by combining document and word semantic embeddings. The centroids of word vectors were considered as topic vectors.

The datasets that has been used to train and test the software is presented in the following section.

III. THE DATASET

The dataset that was used was that of Centralized Public Grievance Redress And Monitoring System (CPGRAMS). The dataset was made available in an online hackathon conducted on the 5th of November 2019. The dataset contained 12 columns and 49,942 rows. The column attributes are, Registration_no, ministry_department, country_name, state_name, dist_name, subject_content, diary_date, closing_date, Source_Name, rating, comments, rating_date. Out of these only ministry_department and subject_content concern our use case. There are 94 departments in total. Each department has complaints ranging from 1 to 20,915.

The dataset is heavily biased towards certain departments but does not focus on some other departments. Due to this reason, we decided to use only those departments that have more than 1000 grievances for classification. The departments that have less than 1000 complaints are clubbed together into one common entity called 'Others'. For better accuracy, only those complaints which have more than 200 characters are considered. Duplicate grievances in the dataset are removed which results in 38,844 unique grievances.

The text was pre-processed by removing punctuation marks, converting lowercase to upper case and numbers to words. The dataset was used to train the supervised machine learning algorithm that was used for identifying the department which should redress the complaint.

Next section presents the proposed model.

IV. THE PROPOSED MODEL

The architecture of the proposed model and the details of implementation is presented in this section.

A. The architecture

The proposed software application provides a user interface to post the grievances in the form of text. To achieve the objectives of identifying the department concerned with redressing the grievance and to provide the concerned department with a report on the gradation of severity of the grievances, the software performs three tasks: They are classification of grievances, keyword extraction and clustering on the basis of keywords. These three tasks are discussed below.

1) *Classification of grievances*: The new grievance submitted to the software application is tagged to a department which is concerned with redressing the grievance. The tagging is performed by a supervised machine learning model which is trained using the data set mentioned in the Section 3. The detailed architecture is shown in the figure 1. After tagging the new grievance to the concerned department, the next step is to extract keywords.

2) *Extraction of keywords*: The objective of the keyword extraction is to identify keywords which best describes the new grievance. The keyword extraction step would return one or more keywords which best describes the grievance. Figure 3 shows the architecture of extraction of keywords.

The next objective is to determine the severity of grievance. Since the grievances are posted by users as text in their own words, it is likely that different words or phrases are used to convey the same grievance. This necessitates the clustering of keywords.

3) *Clustering of keywords*: Clustering is performed on the keywords of grievances which are tagged to a department. For each of the keyword, a set of similar words are generated. The set of similar words are compared with the remaining keywords and if a match is found, then the keyword and similar word are included in a cluster. The process is repeated periodically. Figure 2 shows the architecture of clustering of keywords. The number of members in a cluster is considered as a measure of severity of the grievance.

The details of implementation is presented in the following section.

B. The implementation

The following section presents the details of implementation of each of the three modules namely classification of grievance, extraction of keywords and clustering of keywords. The section presents the algorithms used in each of the modules and also the justification for choosing these algorithms.

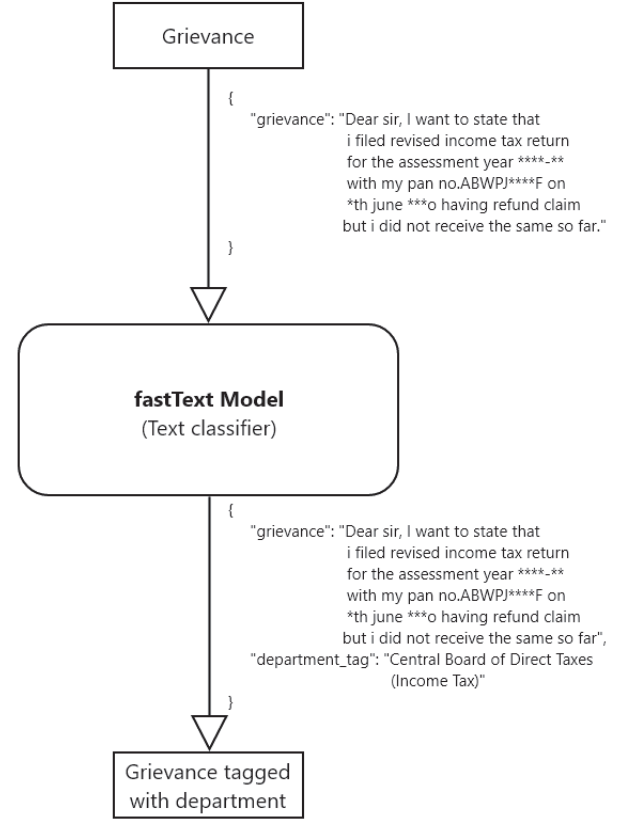


Fig. 1. Classification of grievances

1) *Classification of grievances*: The first step involves determining the department which will redress a particular grievance. In order to determine the model to be used, text classification was performed using algorithms like k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Naive Bayes with One-Hot encoding and TF-IDF scores as a mechanism to vectorize text and fastText. The accuracy of classifying the data set using different algorithms mentioned above are shown in the table I. It can be observed that fastText algorithm gives the best accuracy of classification, and so fastText algorithm is implemented in this application.

2) *Keyword Extraction*: Further, the keywords that best describes the grievance is extracted using keyword extraction algorithms. For this purpose, we make use of keyword extraction using Multipartite graphs and keyword extraction using BERT. Keyword extraction with Multipartite graphs provides

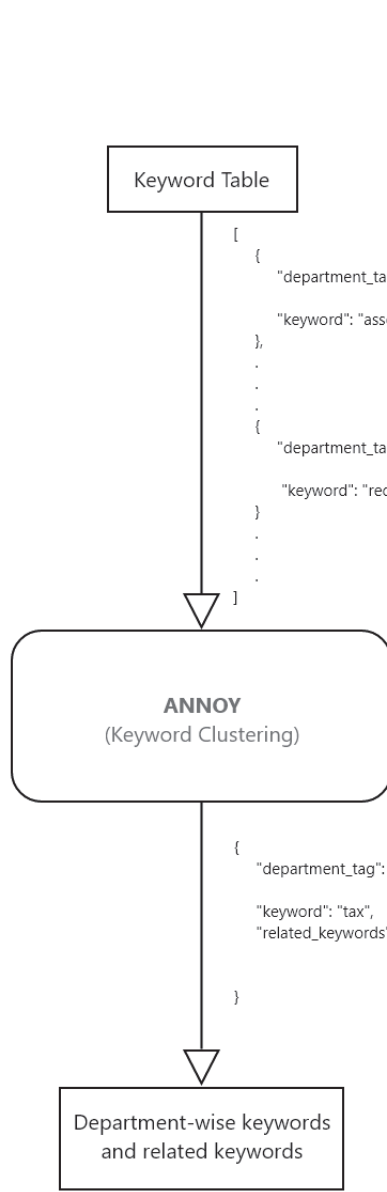


Fig. 2. Clustering of keywords

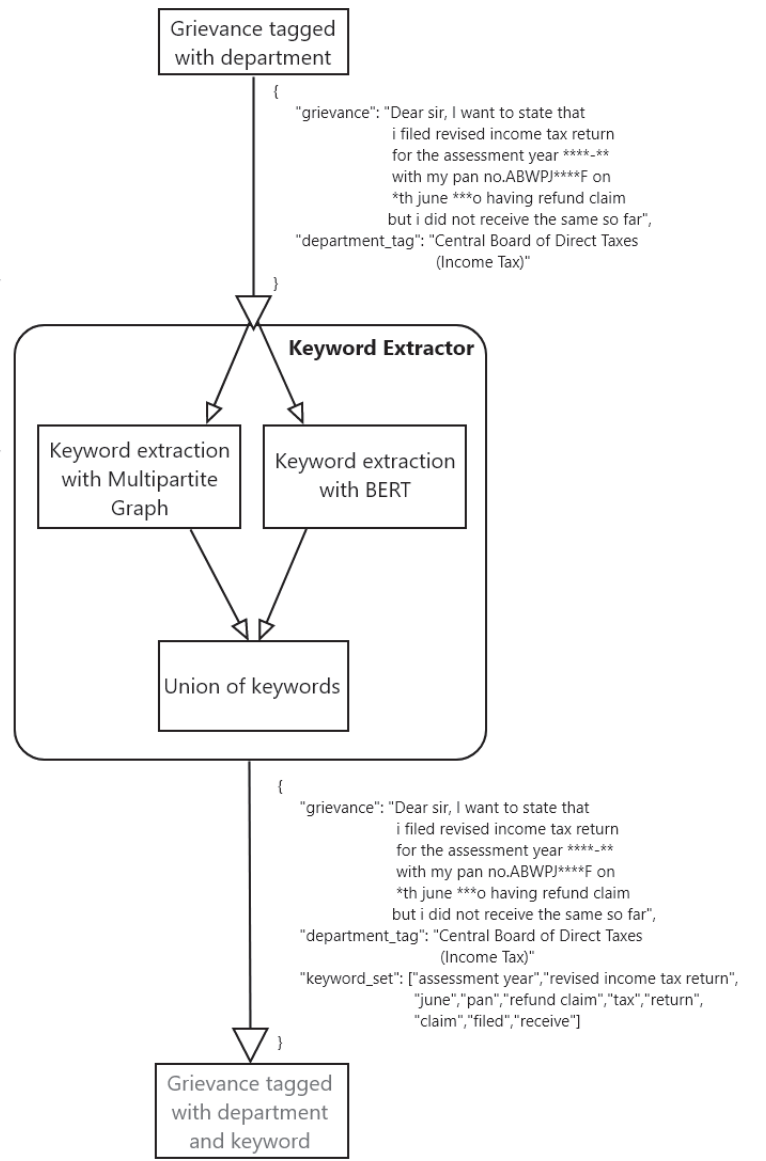


Fig. 3. Extraction of keywords

TABLE I
TEXT CLASSIFICATION ALGORITHM AND THEIR ACCURACY

Algorithm	Accuracy
K-Nearest Neighbors (KNN)	62.65%
Support Vector Machines (SVM)	72.36%
Naive Bayes	72.36%
fastText	90.11%

good n-gram keyphrases where $n \geq 1$, whereas, Keyword extraction with BERT produces excellent 1-gram keyphrases or keywords.

The results of both these algorithms are combined into a set and finally tagged to the grievance from which they were extracted. This result is persisted into a database.

3) *Keyword Clustering*: The keywords summarize the important issues in the grievance. But it is possible that the same grievance is expressed using different texts. Therefore, it is necessary to understand which of these keywords are similar in meaning. As BERT provides excellent vectors that capture the meaning of these keywords, BERT is used to provide word embeddings for each keyword which is clustered using ANNOY. In order to provide a distinct result for each department, distinct clustering models are used for each department. This task is repeated hourly using a Cron job. The main task of these



Fig. 4. Agglomeration of semantically similar complaints

scheduled jobs would be to determine newly found keywords belonging to each department and then pass those keywords to ANNOY models of each department, so that they are retrained on the newly obtained keywords.

After the clusters of all departments are retrained, an array of new related keywords for each keyword of each department is obtained.

Sometimes it is possible that the keywords generated are not completely meaningful due to typographical errors. To avoid such keywords from entering the clustering models we maintain a threshold on the number of mentions (mention of a keyword is the number of grievances to which that particular keyword is tagged) of a keyword within each department, thus ensuring that only meaningful keywords enter the clustering models.

Now that we have one keyword associated with each department and tagged with an array of its related keywords, it becomes possible to determine the most frequent issue by counting the total mentions of a keyword and its related keywords.

V. TEST RESULTS

Keyword extraction and clustering modules were tested using a data set of complaints on a telecom company (COMCAST) [23]. The dataset contained 1841 unique grievances and the application obtained the topics that were most used in the grievances.

Figure 4 shows the topics that were identified as the most frequent among the grievances. For example consider Topic 5. Keywords are `data_limit`, `data_cap` etc., which are all trying to convey the same issue, that is, the users have a problem with the data or Internet usage limits. Similarly, in all the other topics, we can identify the main issue being faced by

the users by simply going through the keywords within that particular topic.

VI. CONCLUSION

This paper presents a design and implementation of a Smart Grievance Redressal System. The software automatically classifies the grievance and maps the grievance to the department concerned with redressing the same. The software can be used by administrations to efficiently manage public grievances. The software identifies frequently used keywords, and uses that information to club similar grievances and helps identify topics and complaints within them, that need to be resolved with priority.

The application can be extended to support analytics on any feedback oriented system. The model can be used to identify trending topics and can also have various other applications such as news analysis, social media topic analysis etc. It is also possible to add support for languages other than English.

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